4. Non-parametric Estimation

The dataset chem in library MASS is made available in R by typing:

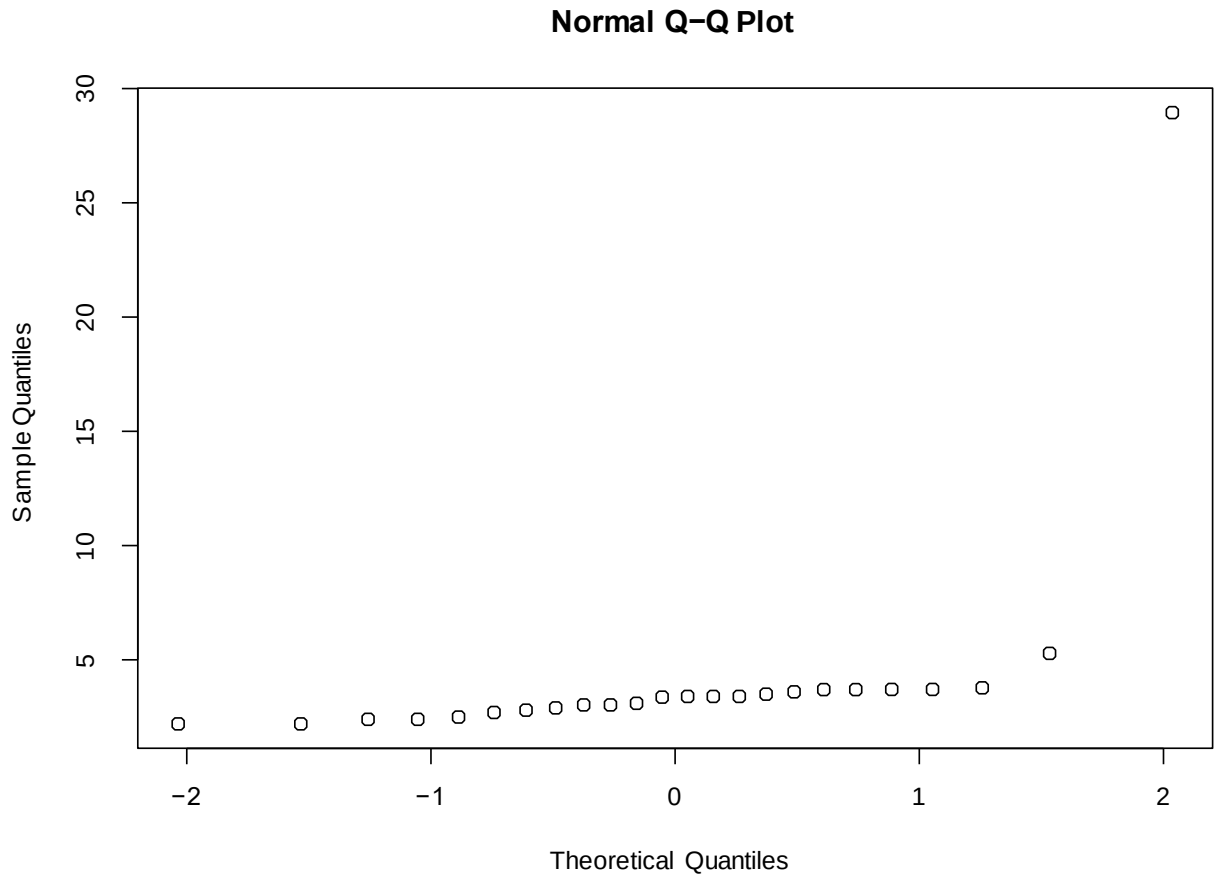
```
> library(MASS)
> data(chem)
```

It contains 24 determinations of copper ($\mu g\, g^{-1}$) in wholemeal flour. We wish to investigate the distribution of these data and obtain confidence intervals for the mean.

2.90	3.10	3.40	3.40	3.70	3.70	2.80	2.50	2.40
2.40	2.70	2.20	5.28	3.37	3.03	3.03	28.95	3.77
3.40	2.20	3.50	3.60	3.70	3.70			

Sample mean = 4.280, sample sd = 5.297

The Q-Q plot reveals an outlier if the data are considered to be normally distributed:

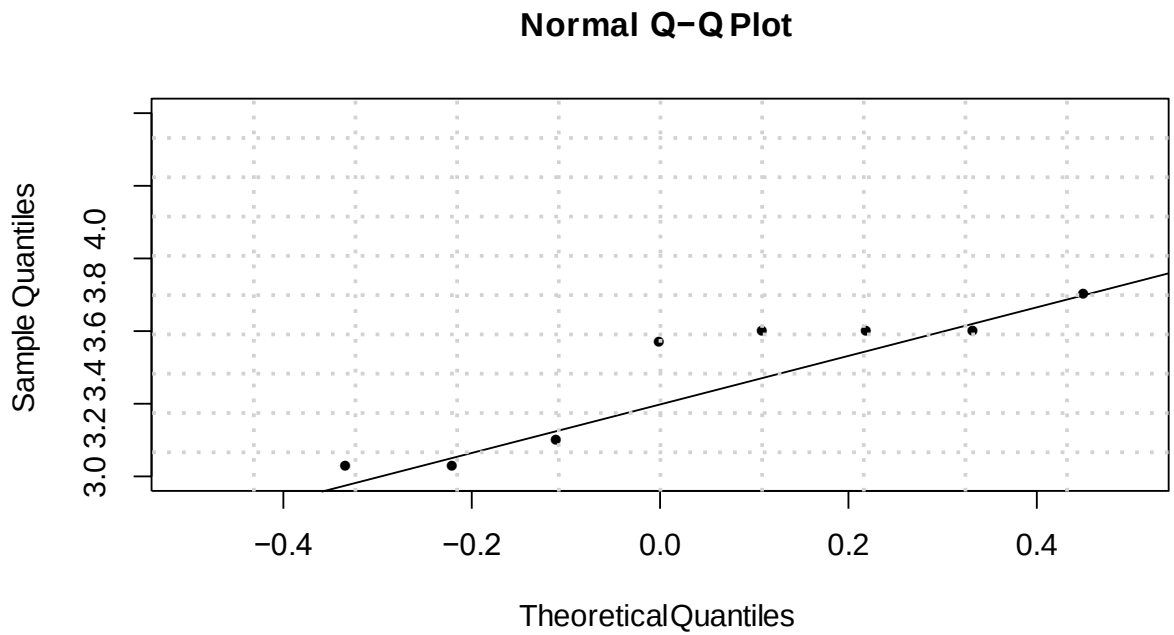


The extreme value, the 17th in the list, can be removed using the command:

```
> chem1=chem[-17]
```

This produces a rather more linear Q-Q plot. So we will now consider the revised data set, chem1.

A zoomed in version of the Q-Q plot is obtained below.



The commands to produce this are:

```
> qqnorm(chem1,xlim=c(-0.5,0.5),ylim=c(3,4))
> qqline(chem1)
> grid(10,10,lwd=2)
```

It can be seen that the intercept is about 3.21. This can be used as an estimate for the population mean.

We can also obtain confidence intervals for the mean or median. Issues to consider:

- Sample size is *not* large for CLT
- Standard deviation σ must be estimated (sample s.d. s)
  this will be reflected in more uncertainty (wider CI)
- If population distn is approximately normal
  we can use t_{n-1} distn for CI
- But, can normality be assumed?
- CI for median can also be obtained,
  but this may also rely on normality assumption.

The data in the chem sample are from a larger study which suggests that the mean copper level is $3.68 \mu g\ g^{-1}$

If we ignore the extreme observation, we can obtain a 95% CI for the mean:

```
> t.test (chem1)
```

giving (2.911, 3.505)

This implies that the data without the extreme point **are not compatible with the larger study**.

However, using the original data CI is (2.044, 6.517) which is compatible. This CI may not be valid –**need to use alternative non-parametric techniques!**

More general:

- Extreme observation adds great uncertainty about what is going on
- It is possible that copper is very heterogeneously distributed
- Much more knowledge of the experimental situation, more data, and alternative methodology are needed